

WEEK - 4

```
Enter password: *****
Current Mongosh Log ID: 6a7ad5d42eadf8251f359489
Connecting to:      mongodb+srv://<credentials>@cluster0.5kk9sey.mongodb.net/
?appName=mongosh+2.9.2
Using MongoDB:      8.0.29 (API Version 1)
Using Mongosh:      2.9.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-56p1t0-shard-0 [primary] test> use studentsdb
switched to db studentsdb
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.insertOne({ name: "A
lice", age: 21, major: "Computer Science" })
{
  acknowledged: true,
  insertedId: ObjectId('6a7ad9032eadf8251f35948a')
}
```

```
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.insertMany([
|   { name: "Alice", age: 21, major: "Computer Science" },
|   { name: "Bob", age: 22, major: "Mathematics" },
|   { name: "Charlie", age: 23, major: "Physics" },
|   { name: "David", age: 21, major: "Computer Science" },
|   { name: "Eve", age: 22, major: "Mathematics" }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a7ad90f2eadf8251f35948b'),
    '1': ObjectId('6a7ad90f2eadf8251f35948c'),
    '2': ObjectId('6a7ad90f2eadf8251f35948d'),
    '3': ObjectId('6a7ad90f2eadf8251f35948e'),
    '4': ObjectId('6a7ad90f2eadf8251f35948f')
  }
}
```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.find()
[
  {
    _id: ObjectId('6a7ad9032eadf8251f35948a'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948b'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948c'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948d'),
    name: 'Charlie',
    age: 23,
    major: 'Physics'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948e'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948f'),
    name: 'Eve',
    age: 22,
    major: 'Mathematics'
  }
]

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.find({
|   major: "Computer Science"
| })
[
  {
    _id: ObjectId('6a7ad9032eadf8251f35948a'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948b'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948e'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  }
]

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.find({
|   name: "Alice"
| })
[
|   {
|     _id: ObjectId('6a7ad9032eadf8251f35948a'),
|     name: 'Alice',
|     age: 21,
|     major: 'Computer Science'
|   },
|   {
|     _id: ObjectId('6a7ad90f2eadf8251f35948b'),
|     name: 'Alice',
|     age: 21,
|     major: 'Computer Science'
|   }
| ]
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.updateOne(
|   { name: "Alice" },
|   { $set: { age: 22 } }
| )
{
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 1,
|   modifiedCount: 1,
|   upsertedCount: 0
| }

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.find({ name: "Alice"
| })
[
|   {
|     _id: ObjectId('6a7ad9032eadf8251f35948a'),
|     name: 'Alice',
|     age: 22,
|     major: 'Computer Science'
|   },
|   {
|     _id: ObjectId('6a7ad90f2eadf8251f35948b'),
|     name: 'Alice',
|     age: 21,
|     major: 'Computer Science'
|   }
| ]

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.deleteOne({
|   name: "Eve"
| })
{ acknowledged: true, deletedCount: 1 }
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.find()
[
  {
    _id: ObjectId('6a7ad9032eadf8251f35948a'),
    name: 'Alice',
    age: 22,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948b'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948c'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948d'),
    name: 'Charlie',
    age: 23,
    major: 'Physics'
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948e'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  }
]

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.createIndex({
|   major: 1
| })
major_1
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { major: 1 }, name: 'major_1' }
]
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.createIndex({
|   major: 1,
|   age: -1
| })
major_1_age_-1
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.dropIndex("major_1")
{
  nIndexesWas: 3,
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1786435955, i: 6 }),
    signature: {
      hash: Binary.createFromBase64('wI1x54xB32q713Ioy4h/CLJqPoQ=', 0),
      keyId: Long('7616462181139218438')
    }
  },
  operationTime: Timestamp({ t: 1786435955, i: 6 })
}

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.aggregate([
| {
|   $group: {
|     _id: "$major",
|     total: { $sum: 1 }
|   }
| })
|
| { _id: 'Mathematics', total: 1 },
| { _id: 'Computer Science', total: 3 },
| { _id: 'Physics', total: 1 }
|
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.aggregate([
| {
|   $group: {
|     _id: "$major",
|     avgAge: { $avg: "$age" }
|   }
| })
|
| { _id: 'Computer Science', avgAge: 21.333333333333332 },
| { _id: 'Physics', avgAge: 23 },
| { _id: 'Mathematics', avgAge: 22 }
|

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.aggregate([
|   {
|     $sort: {
|       age: -1
|     }
|   }
| ])
[
|   {
|     _id: ObjectId('6a7ad90f2eadf8251f35948d'),
|     name: 'Charlie',
|     age: 23,
|     major: 'Physics'
|   },
|   {
|     _id: ObjectId('6a7ad9032eadf8251f35948a'),
|     name: 'Alice',
|     age: 22,
|     major: 'Computer Science'
|   },
|   {
|     _id: ObjectId('6a7ad90f2eadf8251f35948c'),
|     name: 'Bob',
|     age: 22,
|     major: 'Mathematics'
|   },
|   {
|     _id: ObjectId('6a7ad90f2eadf8251f35948b'),
|     name: 'Alice',
|     age: 21,
|     major: 'Computer Science'
|   },
|   {
|     _id: ObjectId('6a7ad90f2eadf8251f35948e'),
|     name: 'David',
|     age: 21,
|     major: 'Computer Science'
|   }
| ]

```

```

Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.courses.insertMany([
|   { course: "DBMS", major: "Computer Science" },
|   { course: "Algebra", major: "Mathematics" },
|   { course: "Quantum Physics", major: "Physics" }
| ])
{
|   acknowledged: true,
|   insertedIds: {
|     '0': ObjectId('6a7ad9902eadf8251f359490'),
|     '1': ObjectId('6a7ad9902eadf8251f359491'),
|     '2': ObjectId('6a7ad9902eadf8251f359492')
|   }
| }

```

```
Atlas atlas-56p1t0-shard-0 [primary] studentsdb> db.students.aggregate([
  {
    $lookup: {
      from: "courses",
      localField: "major",
      foreignField: "major",
      as: "enrolledCourses"
    }
  }
])
[
  {
    _id: ObjectId('6a7ad9032eadf8251f35948a'),
    name: 'Alice',
    age: 22,
    major: 'Computer Science',
    enrolledCourses: [
      {
        _id: ObjectId('6a7ad9902eadf8251f359490'),
        course: 'DBMS',
        major: 'Computer Science'
      }
    ]
  },
  {
    _id: ObjectId('6a7ad90f2eadf8251f35948b'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science',
    enrolledCourses: [
      {
        _id: ObjectId('6a7ad9902eadf8251f359490'),
        course: 'DBMS',
        major: 'Computer Science'
      }
    ]
  },
]
```

```
{
  _id: ObjectId('6a7ad90f2eadf8251f35948c'),
  name: 'Bob',
  age: 22,
  major: 'Mathematics',
  enrolledCourses: [
    {
      _id: ObjectId('6a7ad9902eadf8251f359491'),
      course: 'Algebra',
      major: 'Mathematics'
    }
  ]
},
{
  _id: ObjectId('6a7ad90f2eadf8251f35948d'),
  name: 'Charlie',
  age: 23,
  major: 'Physics',
  enrolledCourses: [
    {
      _id: ObjectId('6a7ad9902eadf8251f359492'),
      course: 'Quantum Physics',
      major: 'Physics'
    }
  ]
},
{
  _id: ObjectId('6a7ad90f2eadf8251f35948e'),
  name: 'David',
  age: 21,
  major: 'Computer Science',
  enrolledCourses: [
    {
      _id: ObjectId('6a7ad9902eadf8251f359490'),
      course: 'DBMS',
      major: 'Computer Science'
    }
  ]
}
]
```